

ASSIGNMENT 2

Due date: Monday, October 12th, 11:30 am.

This assignment counts for 8% of final grade.

Exercise 1

Consider an option to exchange the return of AMZN (Amazon stock) with the return of the SPX (S&P 500) index; this payoff is

$$\left(\frac{S_T^{SPX}}{S_0^{SPX}} - \frac{S_T^{AMZN}}{S_0^{AMZN}} \right)^+$$

The current price of AMZN stock is \$231, and the SPX index is \$6600. The dividend rate for AMZN in the last year is 0%, and for the SPX index is 1.3%. Assume a risk-free rate of $r = 4.8\%$. The implied volatilities provided to you in the Excel spreadsheet correspond to a time-of-expiration $T = 0.288$ (105 days).

Price this option by Monte Carlo simulation: Draw a sample of 10000 points using the Gaussian copula method (use correlation $\rho = 0.5$), map each marginal to the corresponding implied distribution.

Provide the price and the standard error of your simulation in your report.

Exercise 2

Now you are pricing an option on AMZN stock with the same maturity $T = 0.288$, but the payout is $\max(0, K - S_{AMZN}^2)$ with $K = 218^2$. Price this option using the first version of the algorithm described in the slides: Draw a sample of 10000 points from the uniform distribution $\mathcal{U}([0, 1])$ and map each point to the implied distribution for AMZN stock.

Provide the price and the standard error of your simulation in your report.